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# Subject Name: Database Management System

## Week 3

## Module 2 : Relational Model

**Faculty Name : DR UTTAM WAGHMODE**

# Index -


Lecture 7 – ER example & EER MODEL

Lecture 8 –Relational Algebra Operations and Relational algebra queries.

Lecture 9 – ER examples





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# Lecture No: 7: ER example & EER MODEL



## ENTITY

- An **Entity** is a **"thing" or "object"** in the real world that is distinguishable from other objects.
- Entity can be any thing that has an independent existence and about which we collect data. It is also known as **entity type**.
  - Example: *In a School database*, the students, teachers, classes, courses, or projects can be taken as an entity
- Entities are represented by means of **rectangles.**



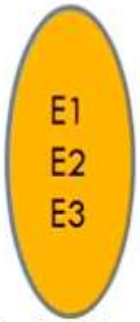

- **Entities** have **attributes** that give them their identity
  - Example: Students have *roll. no., names* and *addresses*
- **Entities** becomes **table** in relational model



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## ENTITY SET

- An **Entity Set** is a collection of similar type of entities, that share same attributes.
  - Example: a *Students set* may contain all the students of a school;
    a *Teachers set* may contain all the teachers of a school from all faculties.



Entity Set

- **Entity sets** need not be **disjoint**
  - Example: the entity set Employee (all employees of a bank) and the entity set Customer (all customers of the bank) may have members in common



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## EXTENDED ER MODEL

- As the complexity of data increased in the late 1980s, it became more and more difficult to use the traditional ER Model for database modelling. Hence some improvements or enhancements were made to the existing ER Model to make it able to handle the complex applications better.
- Hence, as part of the **Extended ER Model**, along with other improvements, three new concepts were added to the existing ER Model:

1. **Generalization**
2. **Specialization**
3. **Aggregation**



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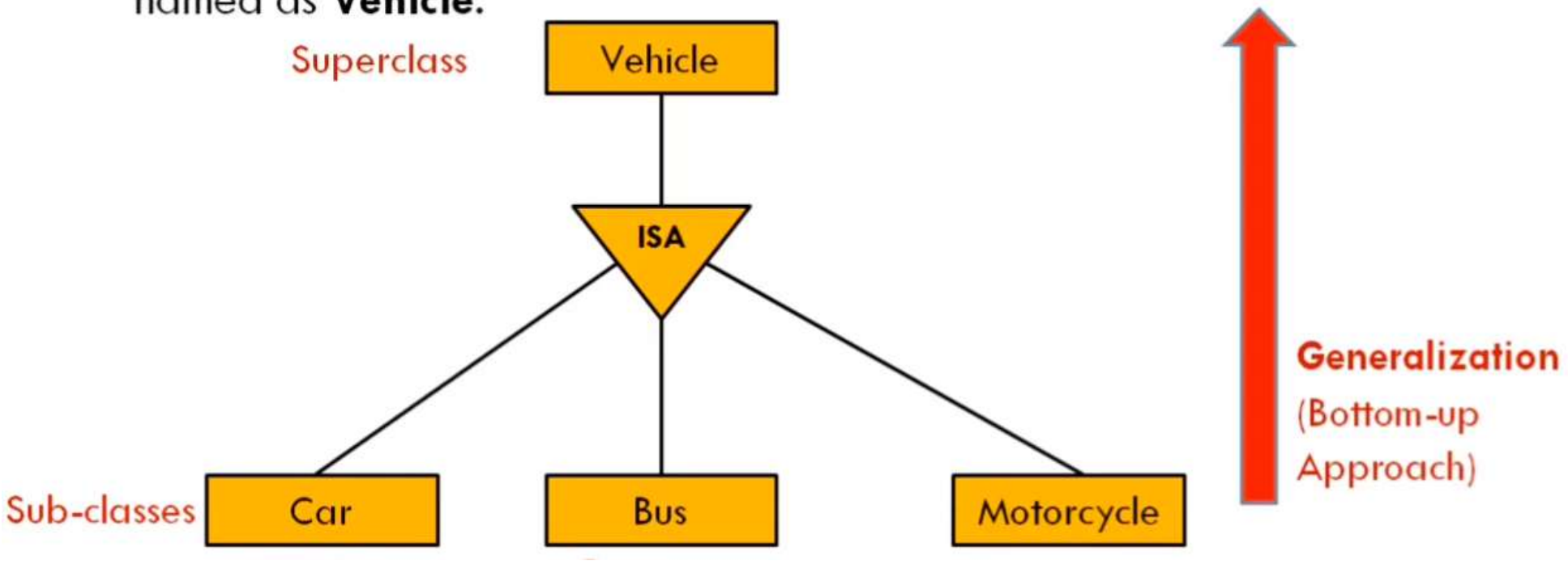
## Generalization:

- **Generalization** is the process of extracting common properties from a set of entities and create a generalized entity from it.
- **Generalization is a "bottom-up approach"** in which two or more entities can be combined to form a higher level entity if they have some attributes in common.
  - subclasses are combined to make a superclass.
- **Generalization is used** to emphasize the similarities among lower-level entity set and to hide differences in the schema



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- Consider we have 3 sub entities Car, Bus and Motorcycle. Now these three entities can be generalized into one higher-level entity (or super class) named as **Vehicle.**






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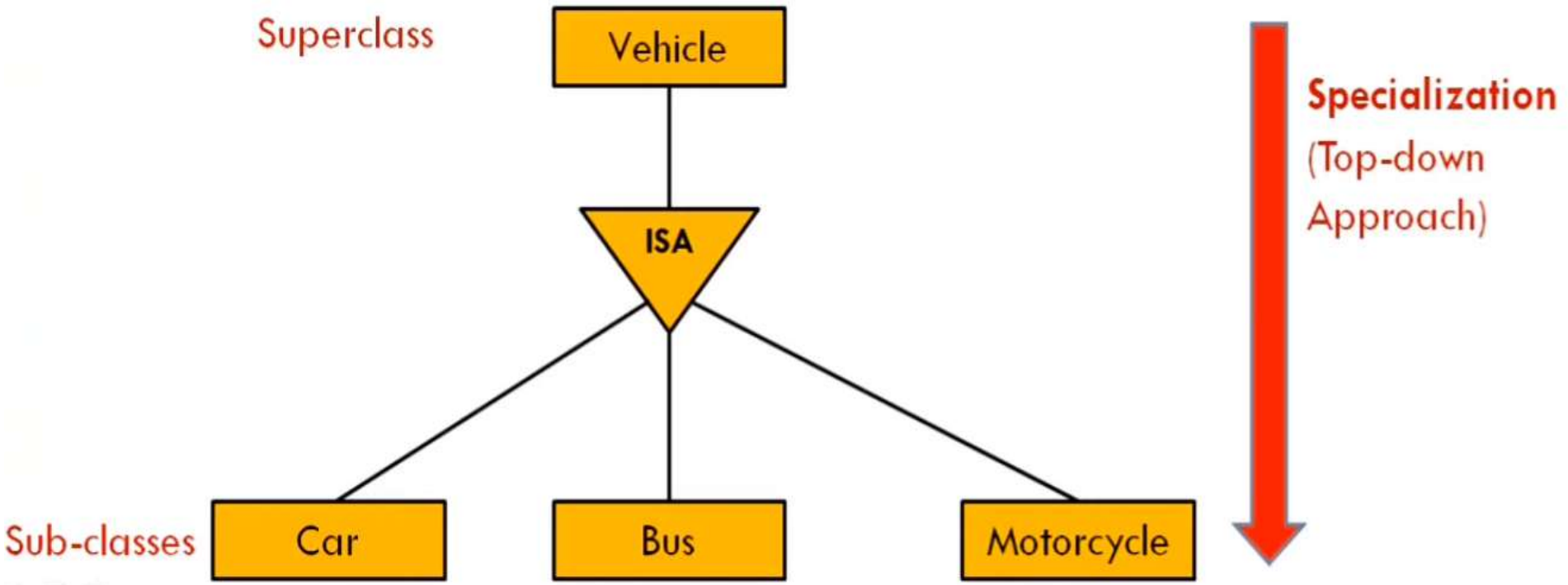
## Specialization:

- Specialization is **opposite** of Generalization
- In **Specialization**, **an entity is broken down into sub-entities based on their characteristics.**
- **Specialization is a "Top-down approach"** where higher level entity is specialized into two or more lower level entities.
- **Specialization is used** to identify the subset of an entity set that shares some distinguishing characteristics.
- **Specialization** can be repeatedly applied to refine the design of schema
- depicted by triangle component labeled **ISA**



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- **Vehicle** entity can be a Car, Truck or Motorcycle.
  - Normally, the superclass is defined first, the subclass and its related attributes are defined next, and relationship set are then added.





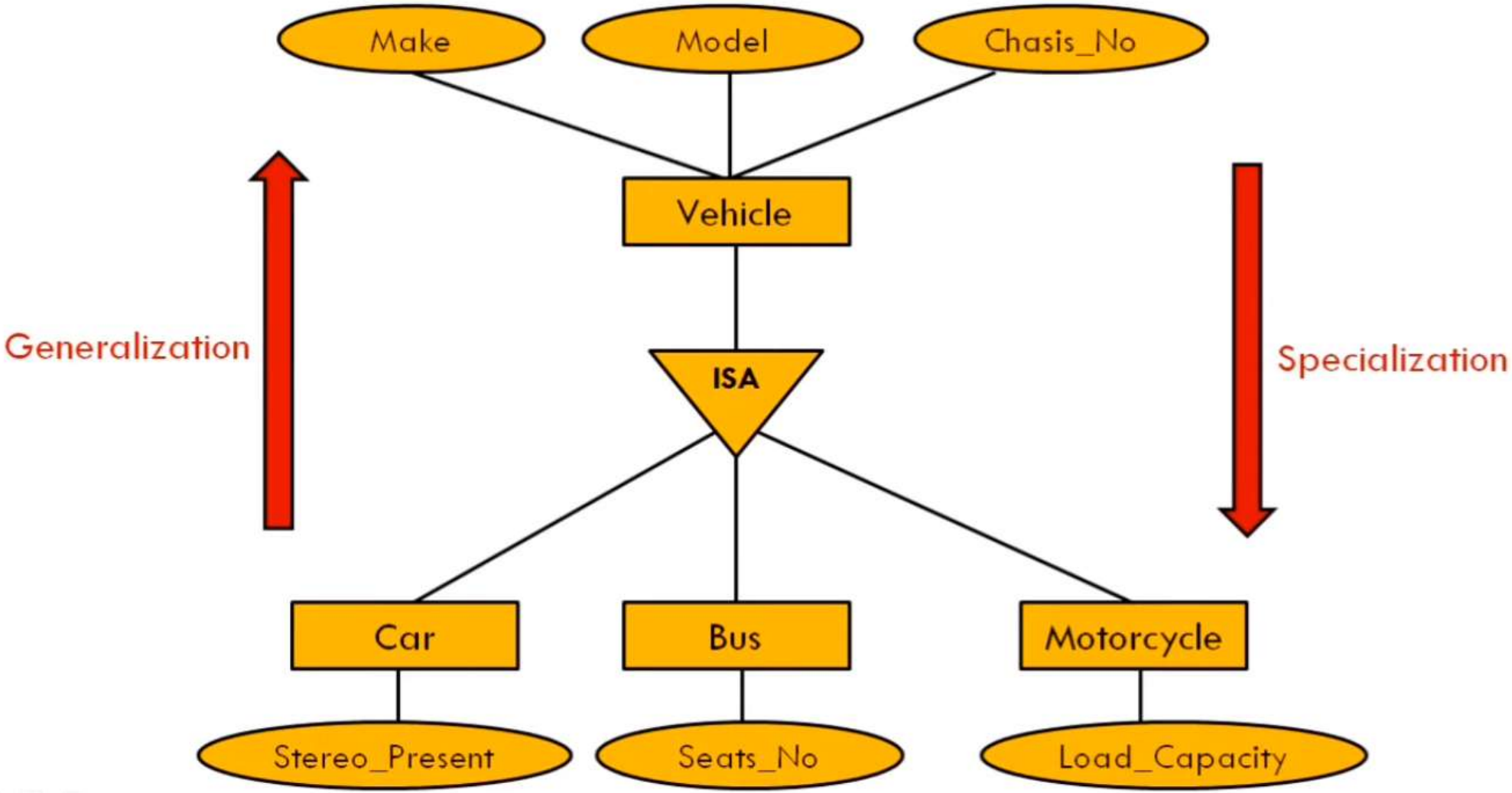

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## Inheritance

- **Inheritance** is an important feature of generalization and specialization.
- **Attribute inheritance allows lower level entities to *inherit the attributes* of higher level entities.**
  - **For example,** Consider relations Car and Bus inheriting the attributes of Vehicle. Thus, Car is described by attributes of super-class Vehicle as well as its own attributes.
- This also extends to **Participation Inheritance** in which ***relationships involving higher-level entity-sets are also inherited by lower-level entity-sets.***
  - A lower-level entity-set can participate in its own relationship-sets, too



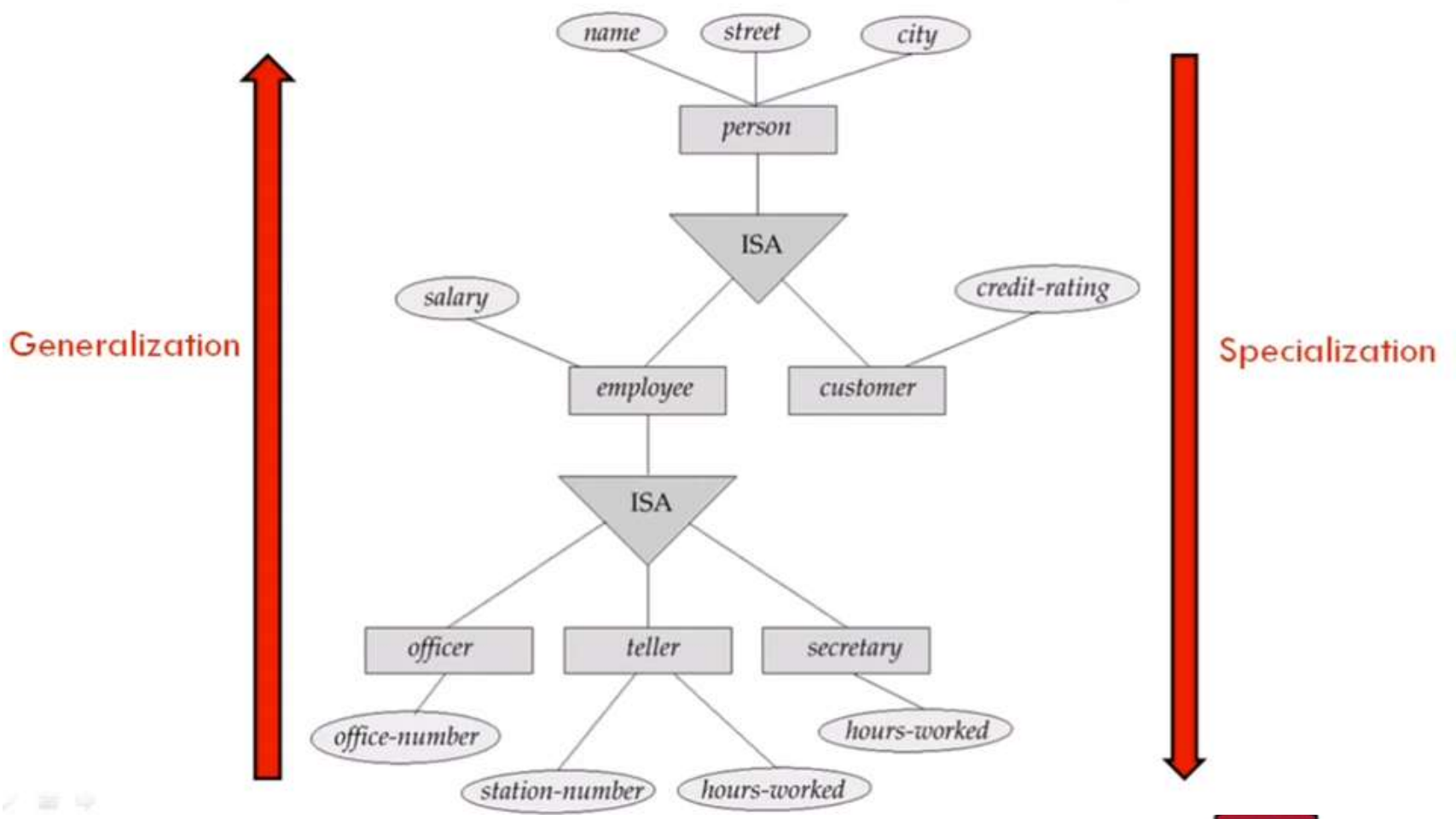
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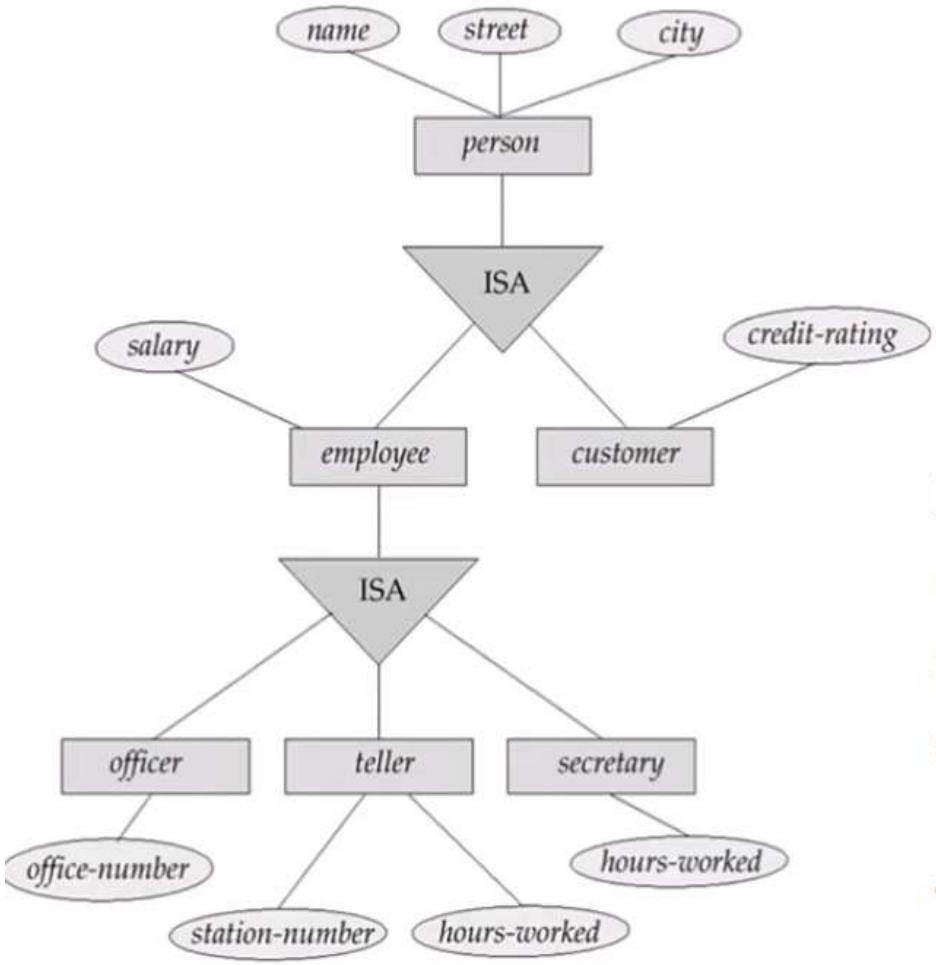
# example






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# How Schema or Tables can be formed?




Four tables can be formed:

1. **customer** (name, street, city, credit_rating)
2. **officer** (name, street, city, salary, office_number)
3. **teller** (name, street, city, salary, station_number, hours_worked)
4. **secretary** (name, street, city, salary, hours_worked)



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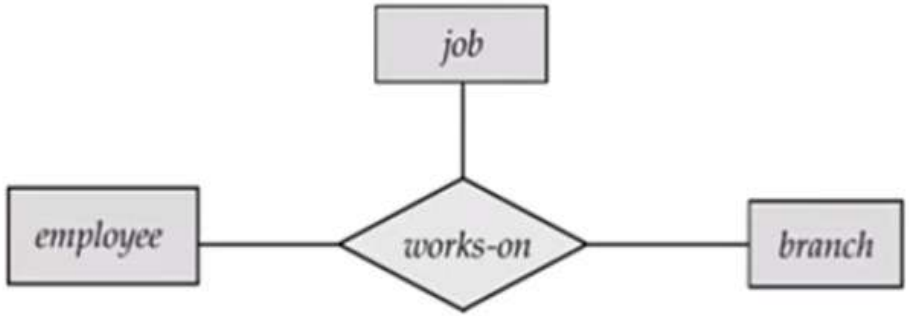
## Aggregation

- **Aggregation** is used when we need to express a **relationship among relationships**
- **Aggregation** is an **abstraction** through which **relationships are treated as higher level entities**
- **Aggregation** is a process when a **relationship between two entities is considered as a single entity** and again this single entity has a **relationship** with **another entity**



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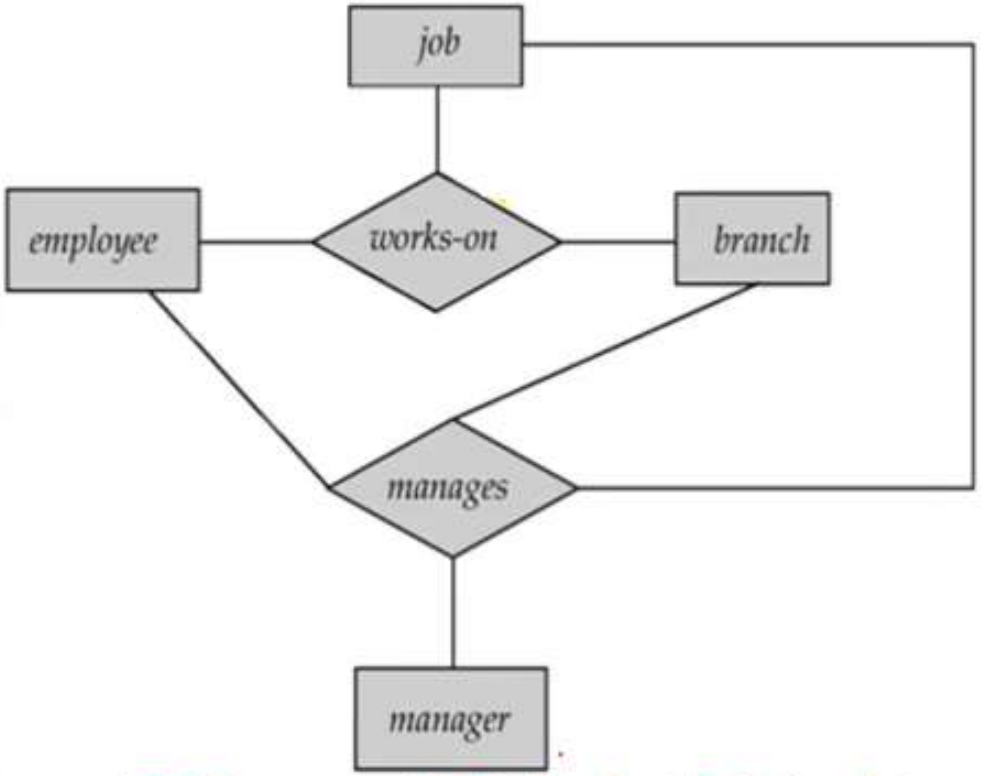
- Basic E-R model can't represent relationships involving other relationships
- Consider a ternary relationship **works_on** between **Employee, Branch and Job.**
  - Am **employee *works on*** a particular **job** at a particular **branch**




- Suppose we want to assign a **manager** for jobs performed by an employee at a branch (i.e. want to assign managers to each employee, job, branch combination)
  - Need a separate manager entity-set
  - Relationship between each manager, employee, branch, and job entity



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ER Diagram with Redundant Relationship

- Relationship sets ***works-on*** and ***manages*** represent overlapping (redundant) information
  - Every *manages* relationship corresponds to a *works-on* relationship
  - However, some *works-on* relationships may not correspond to any *manages* relationships
    - So we can't discard the *works-on* relationship



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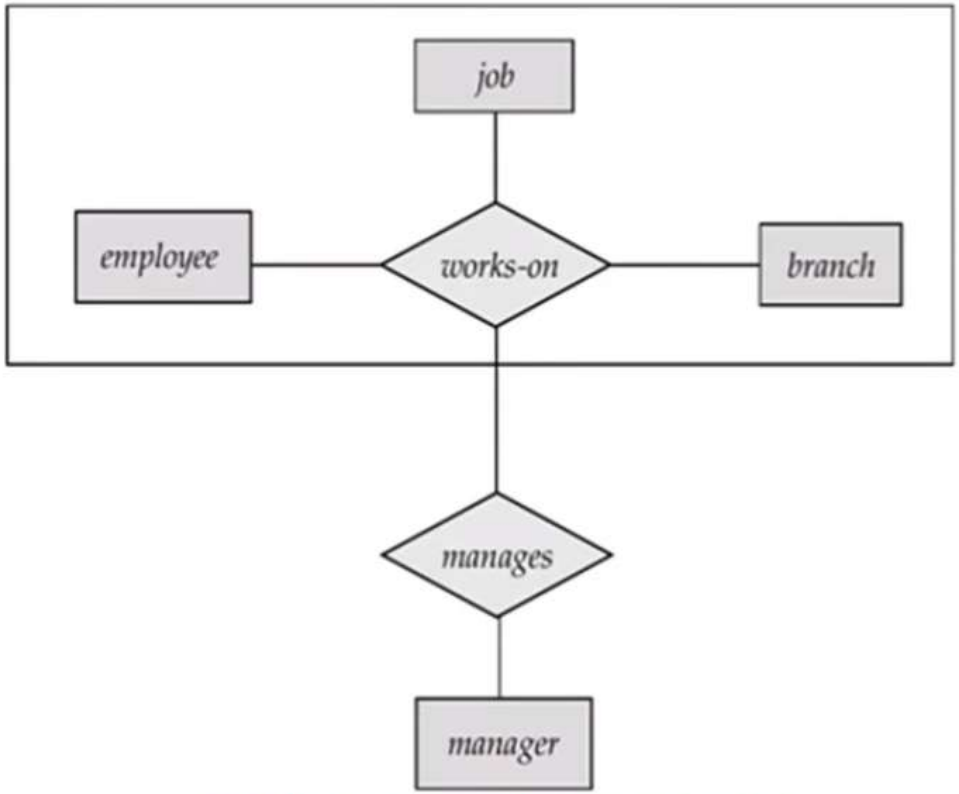
- Eliminate this redundancy via ***aggregation***
  - Treat relationship as an abstract entity
  - Allows relationships between relationships
  - Abstraction of relationship into new entity



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# Example of aggregation

- With **Aggregation** (without introducing redundancy) the ER diagram can be represented as:
  - An employee works on a particular job at a particular branch
  - An employee, branch, job combination may have an associated manager




ER Diagram with Aggregation



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- STUDENT MNAG. SYSTEM:
- https://www.youtube.com/watch?v=4CCkQBn5Fwc

- LIBRARY SYSEM:
- https://www.youtube.com/watch?v=OXpvOx5KjhQ

- ONLINE SHOPPING SYSTEM:
- https://www.youtube.com/watch?v=OPD5JPQdvv4

- MOVIE BOOKING SYSTEM:
- https://www.youtube.com/watch?v=o7ZcrZ-N5Gw&list=PLlGqj2KrYnpSIavv7ZdRkD8VWUwwZ2dF2&index=12



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# **Lecture No: 8:** Relational Algebra Operations and Relational algebra queries.



## Relational algebra

- Introduction to Relational Query Language.
- There are two formal languages used for relational model & those are as follows: Relational Algebra; Relational Calculus.
- The relational Algebra defines a set of operations for the relational Model, & the relational calculus provides higher level declarative language for specifying relational queries.
- The practical language used during database programming is developed later than the relational Algebra & relational Calculas
- Some of the concepts of SQL are based on relational Algebra & relational calculas.



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## A] Relational Algebra-

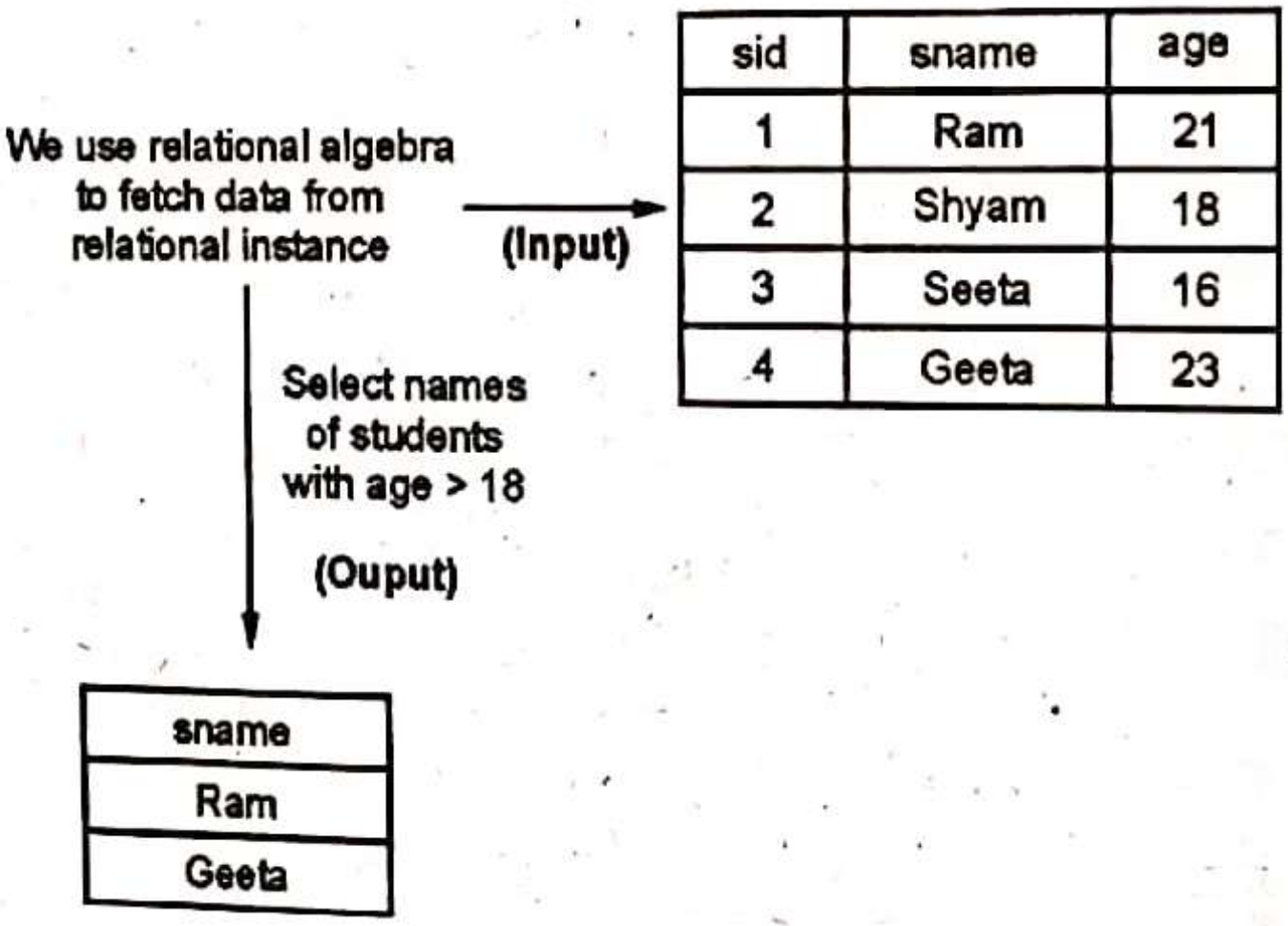
- Defi- Relational Algebra is a procedural Query language which is used to access database tables to retrieve information in the form of relational Algebra expression.
- The queries present in the relational Algebra are denoted using operator's.
- Properties of Relational Algebra :

1. It makes use of operators & use relations (means table) as operands (tables)
2. In relational Algebra, one or two relations are used as input & single relation (table) generated as output without name



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- For eg-






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3] Each relational Algebra is procedural. that means each query describes a set step by step procedure for computing the desired answer, based on the order in which operators are applied in the query.

4] The relational Algebra is based on set theory & it does not allow duplicate entries.

5] It does not make use of any English statements.

➢ Importance of Relational Algebra:

1] it provides formal foundation for relational model operation

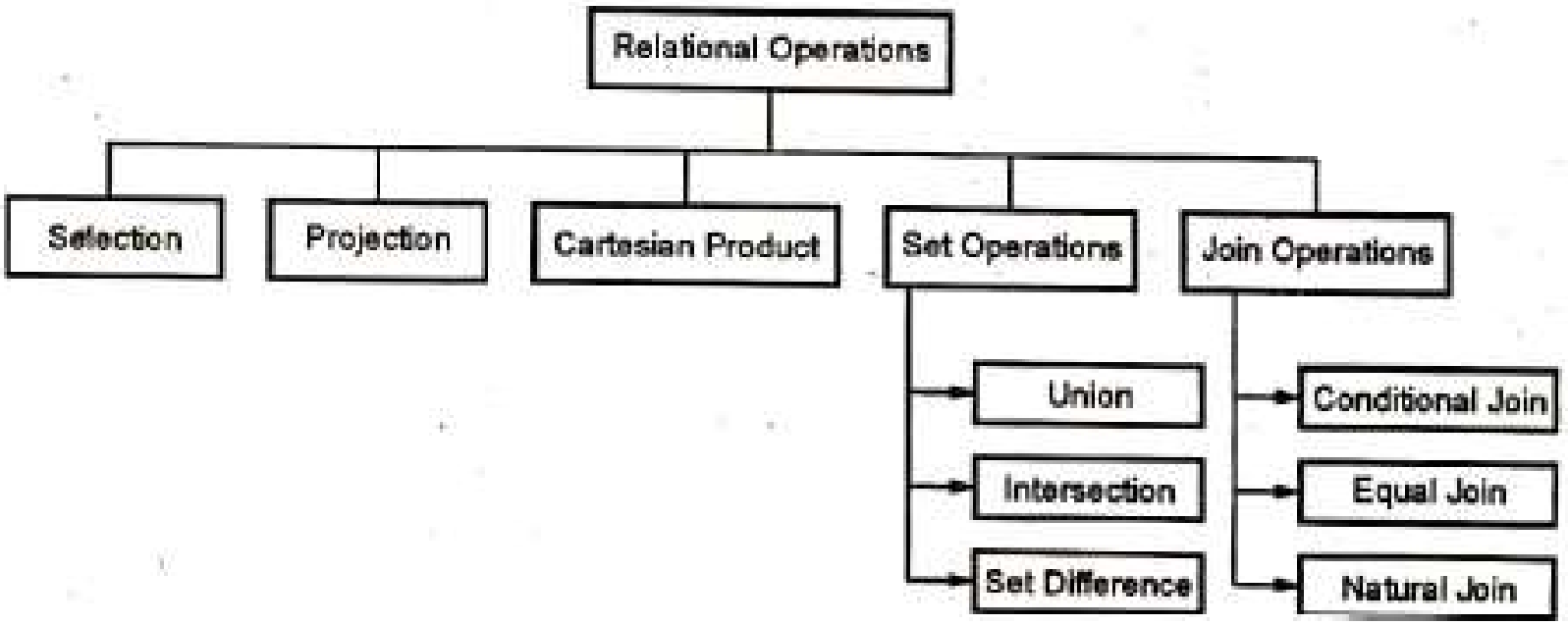
2] it's used as a basic for implementing & optimizing queries in query processing.

3] some of the concepts of relational Algebra are incorporated into SQL



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# Operations in Relational Algebra:






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## A] selection (σ )

- This operation is used to fetch the rows or tuples from table (relation).

Student

| roll_no | name | age | address |
|---|---|---|---|
| 1 | A | 20 | Bhopal |
| 2 | B | 17 | Mumbai |
| 3 | C | 16 | Mumbai |
| 4 | D | 19 | Delhi |
| 5 | E | 18 | Delhi |

**Query 1: Select student whose roll no. is 2**

$$\sigma_{roll_no=2}(Student)$$

| roll_no | name | age | address |
|---|---|---|---|
| 2 | B | 17 | Mumbai |

***Note**: In Selection operation, schema of resulting relation is identical to schema of input relation*



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## The SELECTION operation:

Ex.2: Select the EMPLOYEE tuples whose Department Number is 2 and Salary > 30000.

Sol: $\sigma_{DNo=2 \text{ AND } Salary>30000}(EMPLOYEE)$

| FName | LName | Salary | DNo |
|---|---|---|---|
| Alicia | Smith | 40000 | 2 |

EMPLOYEE

| FName | LName | Salary | DNo |
|---|---|---|---|
| Tom | Ford | 30000 | 3 |
| Amy | Jones | 20000 | 2 |
| Alicia | Smith | 40000 | 2 |
| Frank | Smith | 38000 | 3 |



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Student

| roll_no | name | age | address |
|---|---|---|---|
| 1 | A | 20 | Bhopal |
| 2 | B | 17 | Mumbai |
| 3 | C | 16 | Mumbai |
| 4 | D | 19 | Delhi |
| 5 | E | 18 | Delhi |

**Query 4: Select students whose age is greater than 17 and who lives in Delhi**

$$\sigma_{age > 17 \wedge address = \text{"Delhi"}} (Student)$$

| roll_no | name | age | address |
|---|---|---|---|
| 4 | D | 19 | Delhi |
| 5 | E | 18 | Delhi |



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- Select tuples from a relation "**Books**" where subject is "**database**"

$$\sigma_{\text{subject = "database"}} (\text{Books})$$

- Select tuples from a relation "**Books**" where subject is "**database**" and price is "450"

$$\sigma_{\text{subject = "database"} \wedge \text{price = 450}} (\text{Books})$$

- Select tuples from a relation "**Books**" where subject is "**database**" and price is "**450**" or have a publication **year after 2010**

$$\sigma_{\text{subject = "database"} \wedge \text{price = 450} \vee \text{year > 2010}} (\text{Books})$$



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## B] Projection : Pie(π)

- To display the column in the relation the projector operation is used.

Student

| roll_no | name | age |
|---|---|---|
| 1 | A | 20 |
| 2 | B | 17 |
| 3 | C | 16 |
| 4 | D | 19 |
| 5 | E | 18 |
| 6 | F | 18 |

**Query 1:** Display (or project) the name of students in student table

$$\Pi_{name}(Student)$$



| name |
|---|
| A |
| B |
| C |
| D |
| E |
| F |



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Student

| roll_no | name | age |
|---|---|---|
| 1 | A | 20 |
| 2 | B | 17 |
| 3 | C | 16 |
| 4 | D | 19 |
| 5 | E | 18 |
| 6 | F | 18 |

**Query 2:** Display the roll_no and name of students in student table

$$\Pi_{roll_no,\ name}\ (Student)$$



| roll_no | name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |
| 5 | E |
| 6 | F |



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Student

| roll_no | name | age |
|---|---|---|
| 1 | A | 20 |
| 2 | B | 17 |
| 3 | C | 16 |
| 4 | D | 19 |
| 5 | E | 18 |
| 6 | F | 18 |

**Query 3:** Display the age of students in student table

$$\Pi_{age}(Student)$$



| age |
|---|
| 20 |
| 17 |
| 16 |
| 19 |
| 18 |

**Note**: By default, projection removes duplicate values



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Student

| roll_no | name | age |
|---|---|---|
| 1 | A | 20 |
| 2 | B | 17 |
| 3 | C | 16 |
| 4 | D | 19 |
| 5 | E | 18 |
| 6 | F | 18 |

**Query 4:** Display the roll_no and name of students whose age is greater than 17

$$\Pi_{roll_no,\ name}\ (\sigma_{age\ >\ 17}\ (Student))$$



| roll_no | name |
|---|---|
| 1 | A |
| 4 | D |
| 5 | E |
| 6 | F |



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# Cross product ( X )

- This is use to combine data from two different tables. R1 X R2

**R1**

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 2 |

**R2**

| C | D | E |
|---|---|---|
| $\alpha$ | 10 | a |
| $\beta$ | 10 | a |
| $\beta$ | 20 | b |
| $\gamma$ | 10 | b |



**R1 x R2**

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |



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**R1**

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 2 |

**R2**

| C | D | E |
|---|---|---|
| $\alpha$ | 10 | a |
| $\beta$ | 10 | a |
| $\beta$ | 20 | b |
| $\gamma$ | 10 | b |



**R1 x R2**

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |

R1 x R2
Resulting relation must have

2+3 = 5 attributes
2 x 4= 8 tuples



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- If both input relation have **some attribute having same name**, change the name of the attribute with the name of the relation **"relation_name.attribute_name"**

**R1**

| A | B |
|---|---|
| $\alpha$ | 1 |
| $\beta$ | 2 |

**R2**

| B | D | E |
|---|---|---|
| $\alpha$ | 10 | a |
| $\beta$ | 10 | a |
| $\beta$ | 20 | b |
| $\gamma$ | 10 | b |



**R1 x R2**

| A | R1.B | R2.B | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |





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## Example on multiple operations

- Can build expressions using multiple operations
- **Example:** $\sigma_{A=C}(R1 \times R2)$

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |



**R1 x R2**

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |



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- **Example:** $\prod_{A} (\sigma_{D=20} (R1 \times R2))$

| A |
|---|
| $\alpha$ |
| $\beta$ |



**R1 x R2**

| A | B | C | D | E |
|---|---|---|---|---|
| $\alpha$ | 1 | $\alpha$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 10 | a |
| $\alpha$ | 1 | $\beta$ | 20 | b |
| $\alpha$ | 1 | $\gamma$ | 10 | b |
| $\beta$ | 2 | $\alpha$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 10 | a |
| $\beta$ | 2 | $\beta$ | 20 | b |
| $\beta$ | 2 | $\gamma$ | 10 | b |



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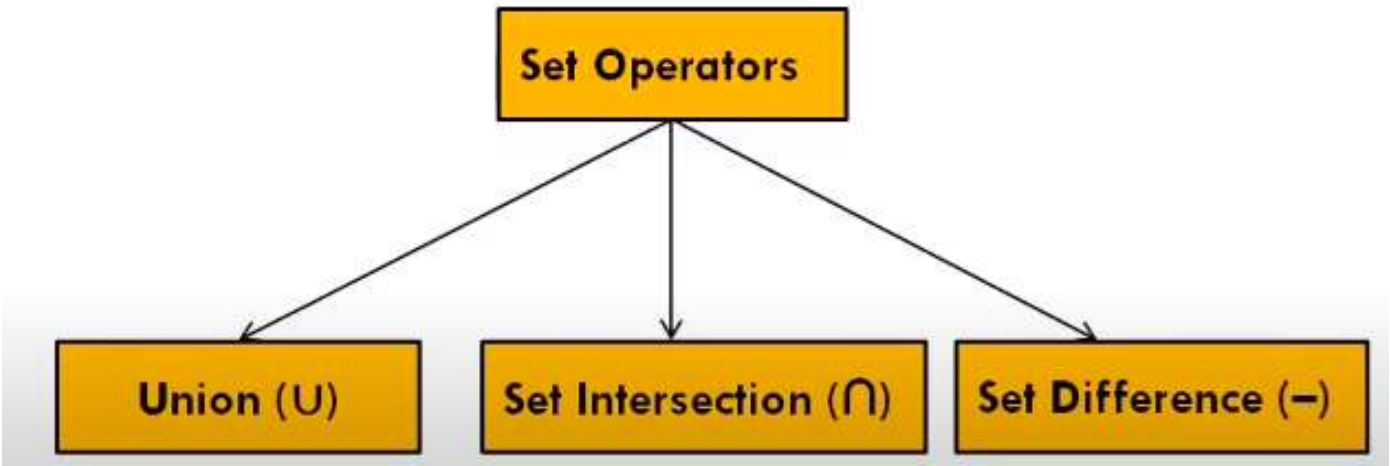
$$\sigma_{author=\text{“Korth”}}(Books \times Articles)$$

**The resulting relation shows all the books and articles written by korth**



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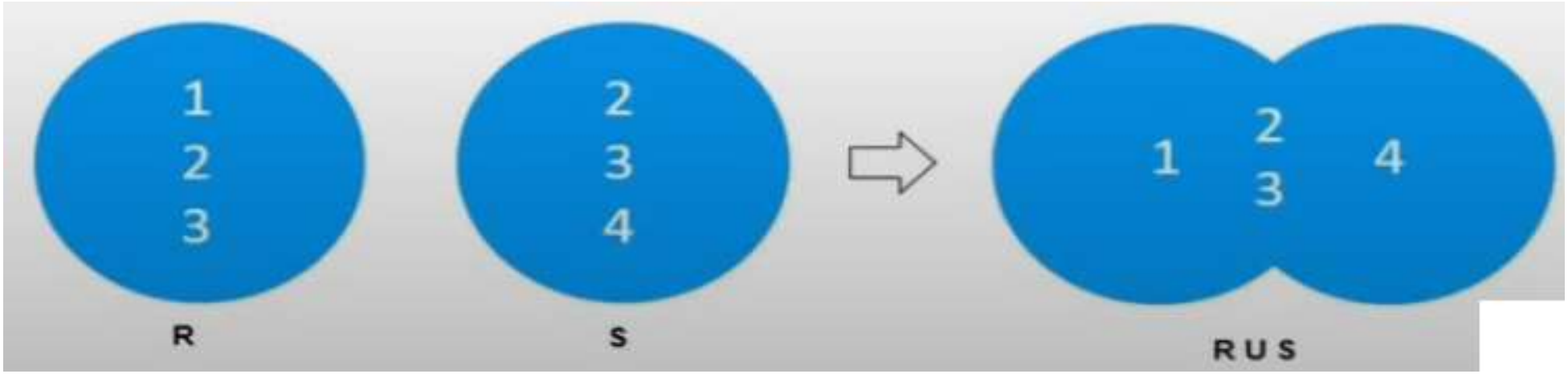
# Set operation






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R U S will be as shown in union operation:






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Student

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Employee

| Emp_no | Name |
|---|---|
| 2 | B |
| 8 | G |
| 9 | H |



(Student) U (Employee)

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |
| 8 | G |
| 9 | H |

**Note: Union is commutative: A U B = B U A**



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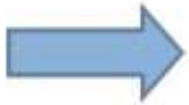
Student

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Employee

| Emp_no | Name |
|---|---|
| 2 | B |
| 8 | G |
| 9 | H |

$\Pi_{Name}$ (Student) U $\Pi_{Name}$ (Employee



| Name |
|---|
| A |
| B |
| C |
| D |
| G |
| H |



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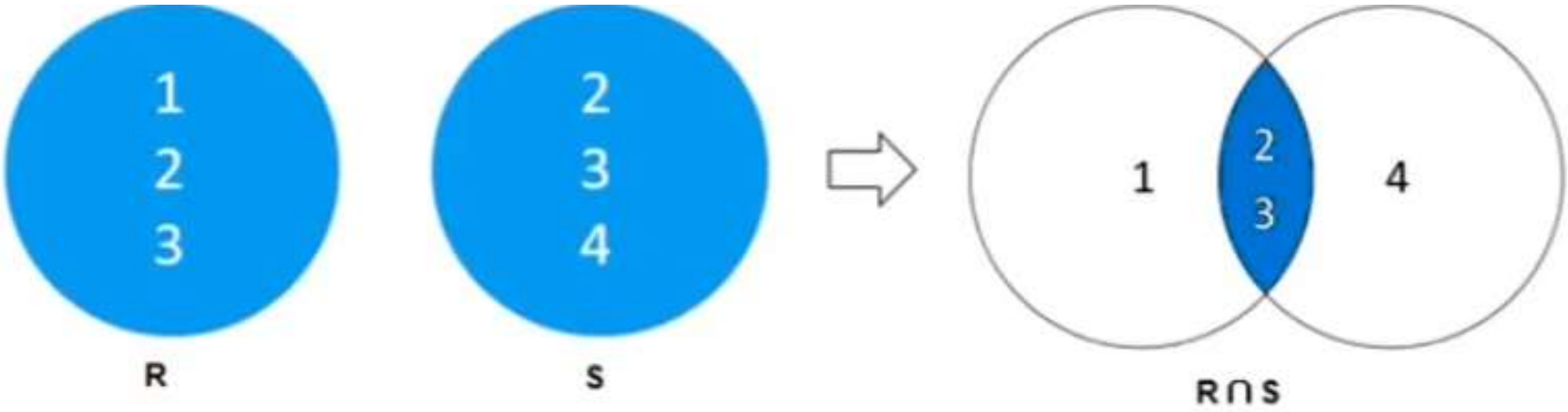
- Find the names of the authors who have either written a book or an article or both.

$$\Pi_{\text{author}}(\text{Books}) \cup \Pi_{\text{author}}(\text{Articles})$$



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# Intersection






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Student

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Employee

| Emp_no | Name |
|---|---|
| 2 | B |
| 8 | G |
| 9 | H |



(Student) ∩ (Employee)

| Roll_no | Name |
|---|---|
| 2 | B |

Student

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Employee

| Emp_no | Name |
|---|---|
| 2 | B |
| 8 | G |
| 9 | H |



$\Pi_{Name}$ (Student) ∩ $\Pi_{Name}$ (Employee)

| Name |
|---|
| B |



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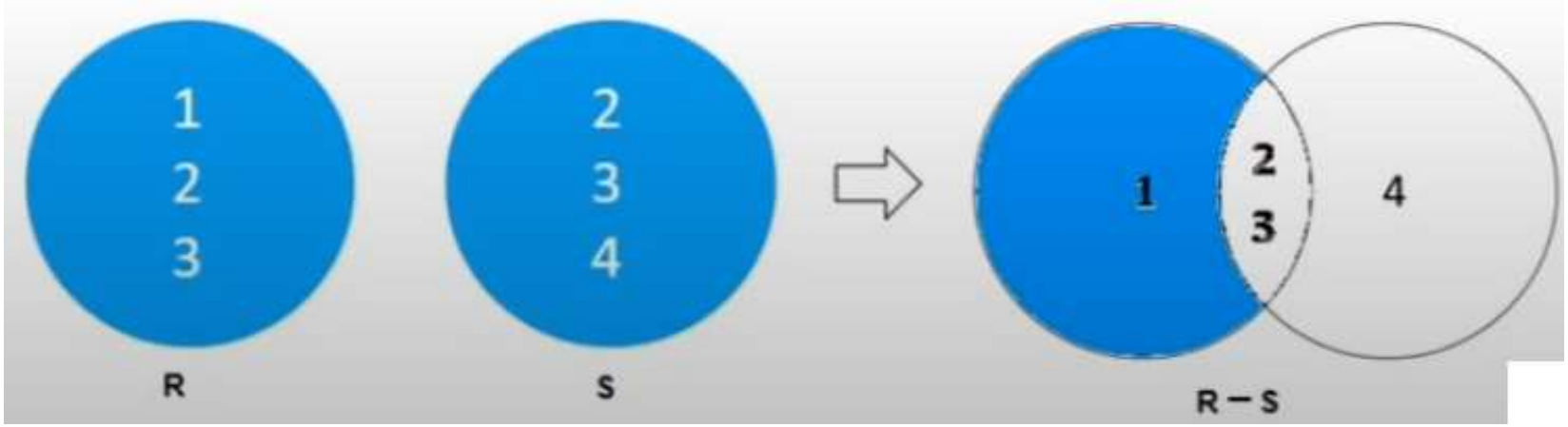
- Find the names of the authors who have written a book and an article both

$$\Pi_{\text{author}}(\text{Books}) \cap \Pi_{\text{author}}(\text{Articles})$$



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# SET DIFFERENCE




Student

| Roll_no | Name |
|---|---|
| 1 | A |
| 2 | B |
| 3 | C |
| 4 | D |

Employee

| Emp_no | Name |
|---|---|
| 2 | B |
| 8 | G |
| 9 | H |

(Student) – (Employee)

| Roll_no | Name |
|---|---|
| 1 | A |
| 3 | C |
| 4 | D |



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**Note**: **1. Set Difference is non-commutative: $A - B \neq B - A$**

**2. $R - (R - S) = R \cap S$**

Intersection can be is derived from set difference that's why intersection ***IS*** derived operator



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## e] Join Operation (⋈ )

- Its used to combine information from 2 or more relations
- formally join can be defined as cross product followed by selection & projections, joins arise much more frequently in practice than plain cross products (X).
- The join operator is used as ⋈
- There are 3 types of joins used in relational algebra.

i) Theta join

ii) Equijoin

iii) Natural join



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## i) Theta join

**S1**

| sid | name | rating | age |
|---|---|---|---|
| 22 | dustin | 7 | 45.0 |
| 31 | lubber | 8 | 55.0 |
| 58 | rusty | 10 | 35.0 |

**R1**

| sid | bid | day |
|---|---|---|
| 22 | 101 | 10/10/96 |
| 58 | 103 | 11/12/96 |

$$S1 \bowtie_{S1.sid < R1.sid} R1$$

| S1.Sid | sname | rating | age | R1.sid | bid | day |
|---|---|---|---|---|---|---|
| 22 | dustin | 7 | 45.0 | 58 | 103 | 11/12/96 |
| 31 | lubber | 8 | 55.0 | 58 | 103 | 11/12/96 |



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$S1 \bowtie_{S1.sid < R1.sid} R1$ is

**Equivalent to:** $\sigma_{S1.sid < R1.sid}(S1 \times R1)$

**WE CAN WRITE IT AS,**

$$\text{So, } A \bowtie_{\theta} B = \sigma_{\theta}(A \times B)$$



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## EQUIJOIN :

S1

| sid | name | rating | age |
|---|---|---|---|
| 22 | dustin | 7 | 45.0 |
| 31 | lubber | 8 | 55.0 |
| 58 | rusty | 10 | 35.0 |

R1

| sid | bid | day |
|---|---|---|
| 22 | 101 | 10/10/96 |
| 58 | 103 | 11/12/96 |

$$S1 \bowtie_{S1.sid = R1.sid} R1$$

| S1.Sid | sname | rating | age | R1.sid | bid | day |
|---|---|---|---|---|---|---|
| 22 | dustin | 7 | 45.0 | 22 | 101 | 10/10/96 |
| 58 | rusty | 10 | 35.0 | 58 | 103 | 11/12/96 |

Equivalent to $\sigma_{S1.sid = R1.sid}(S1 \times R1)$



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## EQUIJOIN

**DIFFERENT ATTRIBUTES NAMES BUT SAME DOMAIN**

**Student**

| SID | Name | Std |
|---|---|---|
| 101 | Rohan | 11 |
| 102 | Mira | 12 |

**Subjects**

| Class | Subject |
|---|---|
| 11 | Maths |
| 11 | Physics |
| 12 | English |
| 12 | Chemistry |

$$\text{Student} \bowtie_{\text{Student.Std} = \text{Subjects.Class}} \text{Subjects}$$

| SID | Name | Std | Class | Subject |
|---|---|---|---|---|
| 101 | Rohan | 11 | 11 | Maths |
| 101 | Rohan | 11 | 11 | Physics |
| 102 | Mira | 12 | 12 | English |
| 102 | Mira | 12 | 12 | Chemis |



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# NATURAL JOIN (*)

r = (A, B, C, D)  s = (B, D, E)

- Resulting schema of r ⋈ s = (A, B, C, D, E)
- **r ⋈ s** is defined as:

$$\Pi_{r.A,\, r.B,\, r.C,\, r.D,\, s.E} (\sigma_{r.B = s.B \,\wedge\, r.D = s.D} (r \times s))$$

**r**

| A | B | C | D |
|---|---|---|---|
| α | 1 | α | a |
| β | 2 | γ | a |
| γ | 4 | β | b |
| α | 1 | γ | a |
| δ | 2 | β | B |

**s**

| B | D | E |
|---|---|---|
| 1 | a | α |
| 3 | a | β |
| 1 | a | γ |
| 2 | b | δ |
| 3 | b | ϵ |



**r ⋈ s**

| A | B | C | D | E |
|---|---|---|---|---|
| α | 1 | α | a | α |
| α | 1 | α | a | γ |
| α | 1 | γ | a | α |
| α | 1 | γ | a | γ |
| δ | 2 | β | b | δ |



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# NATURAL JOIN

**Courses**

| CID | Course | Dept |
|---|---|---|
| CS01 | Database | CS |
| ME01 | Mechanics | ME |
| EE01 | Electronics | EE |

**HOD**

| Dept | Head |
|---|---|
| CS | Rohan |
| ME | Sara |
| EE | Jiya |

**Courses ⋈ HOD**

| CID | Course | Dept | Head |
|---|---|---|---|
| CS01 | Database | CS | Rohan |
| ME01 | Mechanics | ME | Sara |
| EE01 | Electronics | EE | Jiya |

$$\Pi_{CID,\ Course,\ Courses.Dept,\ Head}\ (\sigma_{Courses.Dept\ =\ HOD.Dept}(Courses\ x\ HOD))$$



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# NATURAL JOIN

S1

| sid | name | rating | age |
|---|---|---|---|
| 22 | dustin | 7 | 45.0 |
| 31 | lubber | 8 | 55.0 |
| 58 | rusty | 10 | 35.0 |

R1

| sid | bid | day |
|---|---|---|
| 22 | 101 | 10/10/96 |
| 58 | 103 | 11/12/96 |

S1 ⋈ R1

| sid | sname | rating | age | bid | day |
|---|---|---|---|---|---|
| 22 | dustin | 7 | 45.0 | 101 | 10/10/96 |
| 58 | rusty | 10 | 35.0 | 103 | 11/12/96 |



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# NATURAL JOIN

**Courses**

| CID | Course |
|---|---|
| 100 | Database |
| 101 | Mechanics |
| 102 | Electronics |

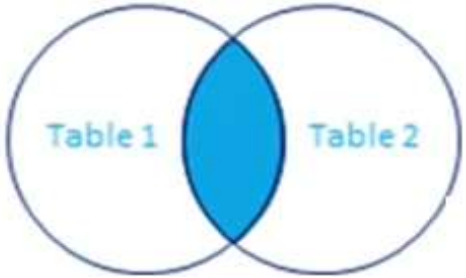
**HOD**

| CID | Name |
|---|---|
| 100 | Rohan |
| 102 | Sara |
| 104 | Jiya |

**Courses ⋈ HOD**

| CID | Course | Name |
|---|---|---|
| 100 | Database | Rohan |
| 102 | Electronics | Sara |

**Inner Join**






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## OUTER JOIN

- **Outer Join = Natural Join + Extra information**

  (from left table, right table or both table)

- There are **three kinds of outer joins:**

  1. **Left outer join**
  2. **Right outer join**
  3. **Full outer join**

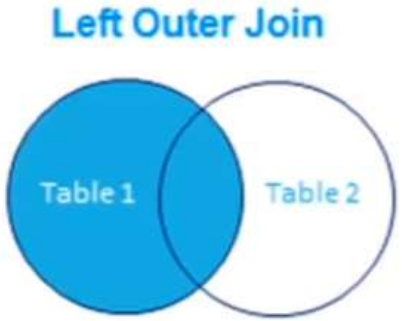


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# LEFT OUTER JOIN

When applying **join** on two relations R1 and R2, some tuples of R1 or R2 does not appear in result set which does not satisfy the join conditions. **But..**

- In **Left outer join**, ***all the tuples from the Left relation R1 are included in the resulting relation.*** The tuples of R1 which do not satisfy join condition will have values as **NULL** for attributes of R2.
- In short:
  - **All** record from **left** table
  - Only matching records from right table
- **Symbol**: ⟕
- **Notation**: **R1 ⟕ R2**






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## LEFTOUTER JOIN

**Courses**

| CID | Course |
|---|---|
| 100 | Database |
| 101 | Mechanics |
| 102 | Electronics |

**HOD**

| CID | Name |
|---|---|
| 100 | Rohan |
| 102 | Sara |
| 104 | Jiya |

**Courses ⟕ HOD**

| CID | Course | Name |
|---|---|---|
| 100 | Database | Rohan |
| 101 | Mechanics | **NULL** ✓ |
| 102 | Electronics | Sara |



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## RIGHT OUTER JOIN

- In **Right outer join**, ***all the tuples from the right relation R2 are included in the resulting relation.*** The tuples of R2 which do not satisfy join condition will have values as **NULL** for attributes of R1.
- In short:
  - **All** record from **right** table
  - Only matching records from left table
- **Symbol**: ⟖
- **Notation**: **R1 ⟖ R2**






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**Courses**

| CID | Course |
|---|---|
| 100 | Database |
| 101 | Mechanics |
| 102 | Electronics |

**HOD**

| CID | Name |
|---|---|
| 100 | Rohan |
| 102 | Sara |
| 104 | Jiya |

**Courses ⋈ HOD**

| CID | Course | Name |
|---|---|---|
| 100 | Database | Rohan |
| 102 | Electronics | Sara |
| 104 | **NULL** | Jiya |



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## FULL OUTER JOIN

**Courses**

| CID | Course |
|---|---|
| 100 | Database |
| 101 | Mechanics |
| 102 | Electronics |

**HOD**

| CID | Name |
|---|---|
| 100 | Rohan |
| 102 | Sara |
| 104 | Jiya |

**Courses ⟗ HOD**

| CID | Course | Name |
|---|---|---|
| 100 | Database | Rohan |
| 101 | Mechanics | **NULL** |
| 102 | Electronics | Sara |
| 104 | **NULL** | Jiya |



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# Rename (ρ )

- *Suppose we want to do cartesian product between same table then* ***one of the table should be renamed with another name***
- ***R x R***

  *(Ambiguity will be there)*

***R***

| A | B |
|---|---|
| α | 1 |
| β | 2 |

***R x R***

| R.A | R.B | R.A | R.B |
|---|---|---|---|
| α | 1 | α | 1 |
| α | 1 | β | 2 |
| β | 2 | α | 1 |
| β | 2 | β | 2 |

- ***R x $\rho_S$ (R)***

  ***(Rename R to S)***

***R x $\rho_S$ (R)***

| R.A | R.B | S.A | S.B |
|---|---|---|---|
| α | 1 | α | 1 |
| α | 1 | β | 2 |
| β | 2 | α | 1 |
| β | 2 | β | 2 |

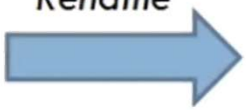


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- **Example-1:** Query to find the female students from Student relation and rename the relation Student as FemaleStudent and the attributes of Student - RollNo, SName as Sno, Name.

***Student***

| RollNo | SName | Gender |
|---|---|---|
| 1 | Neha | F |
| 2 | Suman | F |
| 3 | Sohan | M |
| 4 | Mohan | M |
| 5 | Rohan | M |




***FemaleStudent***

| SNo | Name |
|---|---|
| 1 | Neha |
| 2 | Suman |

$$\rho_{\text{FemaleStudent(Sno, Name)}}\left(\pi_{\text{RollNo, SName}}\left(\sigma_{\text{Gender='F'}}(\text{Student})\right)\right)$$



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**Division operator can be applied if and only if:**

- Attributes of B is **proper subset** of Attributes of A.
- The relation returned by **division operator** will have **attributes =(All attributes of A – All attributes of B).**
- The relation returned by **division operator** will **return** those **tuples from relation A which are associated to every B's tuple**.

**Example 1:**

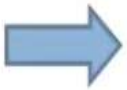
**A**

| x | y |
|---|---|
| a | 1 |
| b | 2 |
| a | 2 |
| d | 4 |

÷

**B**

| y |
|---|
| 1 |
| 2 |

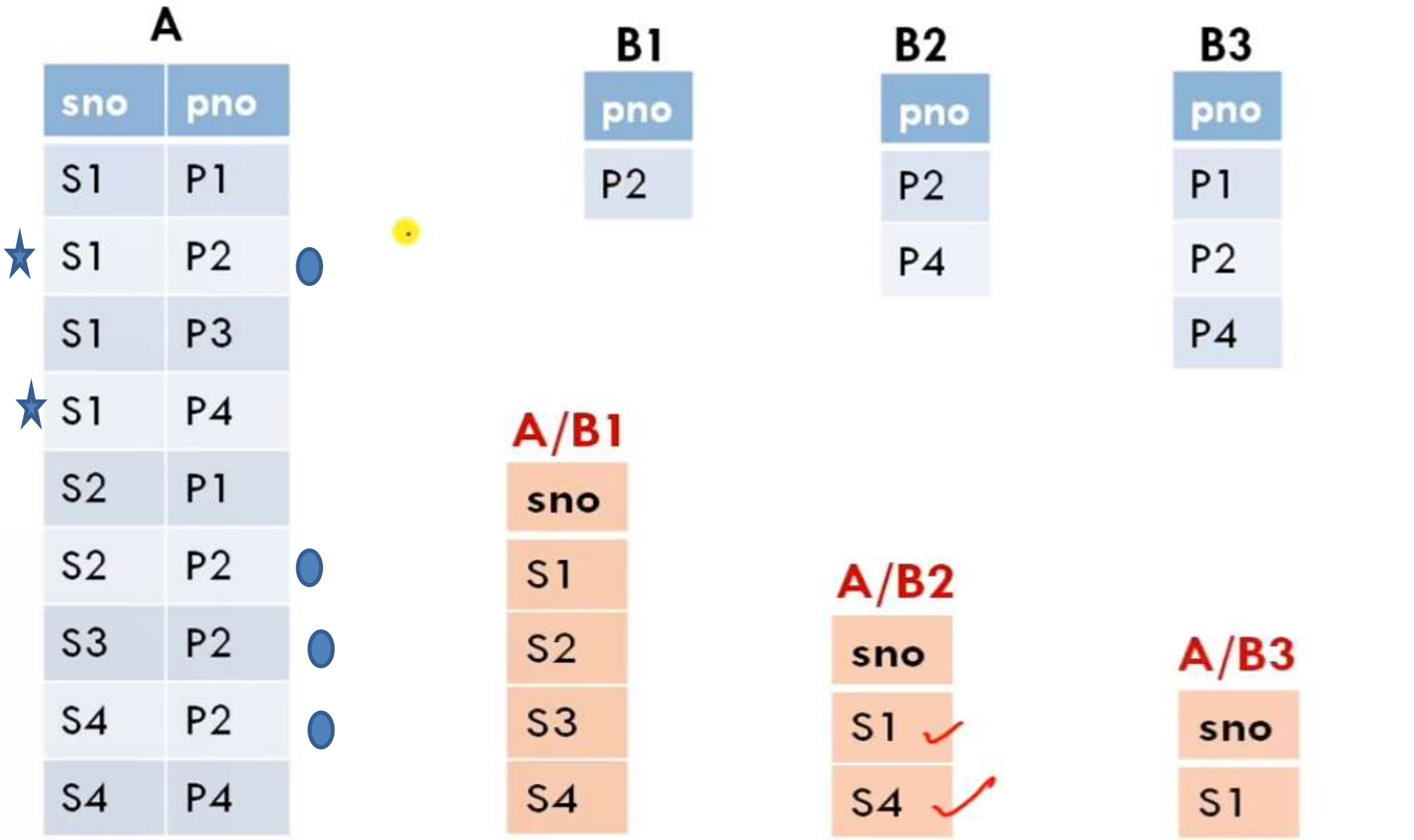


**A÷B**

| x |
|---|
| a |



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# Lecture No: 9:ER examples



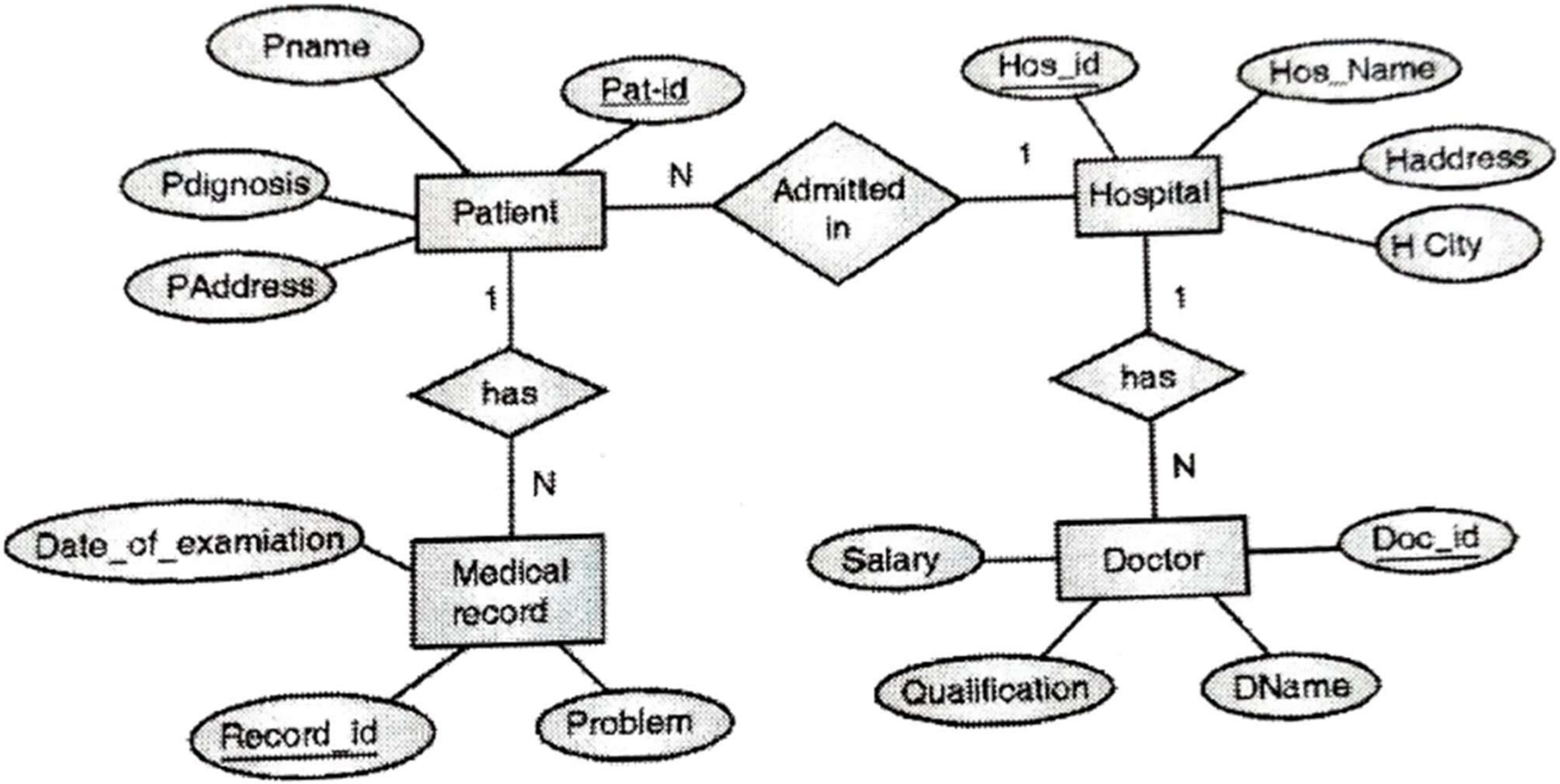
## ER examples

- Draw ER for following: Construct an ER diagram for a hospital with a set of patients and the set of medical doctors associated with each patient, a record of various test and examination conducted.



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# ER examples






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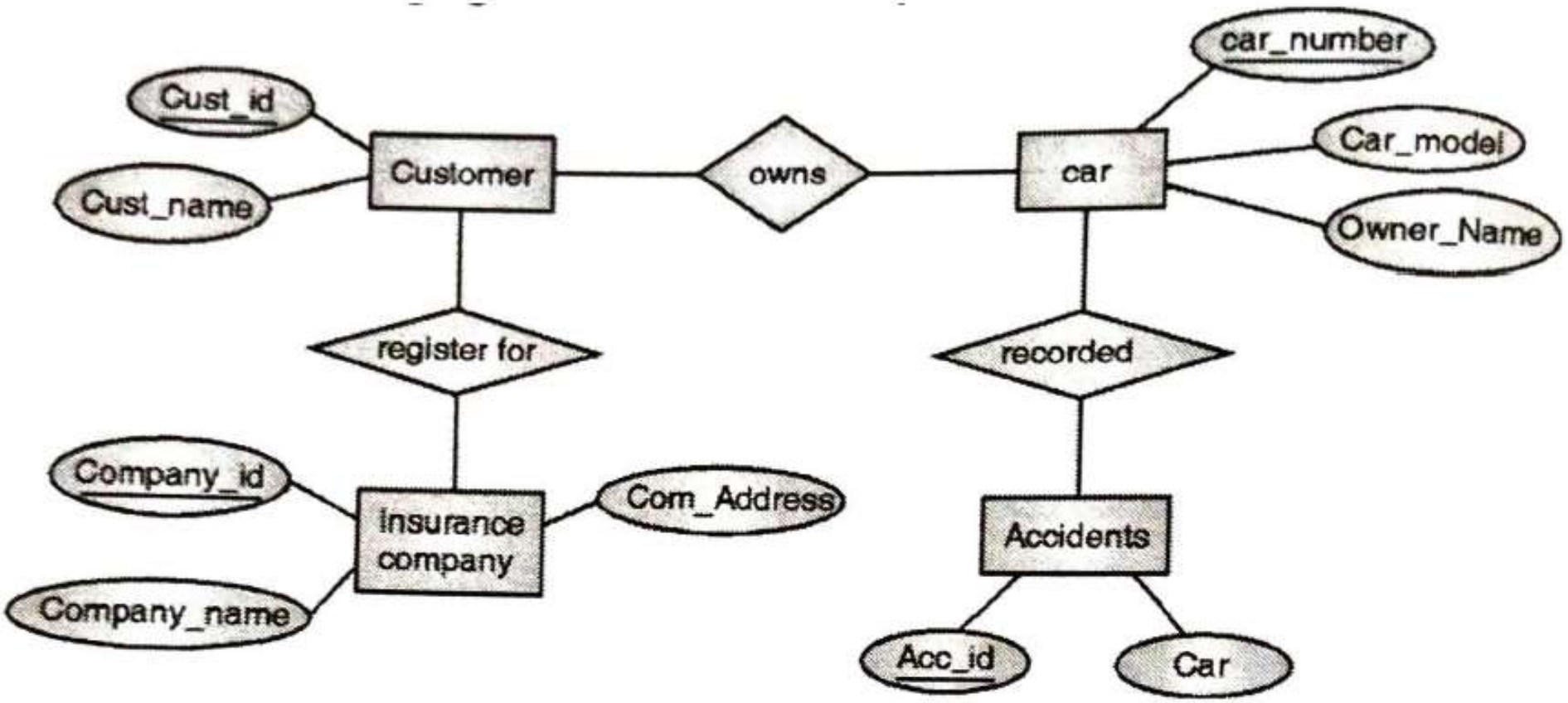
## ER examples

- Construct an ER diagram for a car insurance company agency that has a set of customers each of whom owns one or more cars. Each car has associated with it zero or any number of recorded accidents.



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# ER examples






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# Thank You



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